

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Name:-Sapna Prasad	Roll No:-29
Date of Assignment:-12/08/2026	Date of submission:-12/08/2026

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

Code:-

```
Parc no 8.py - C:/Users/info/OneDrive/Documents/Sapna DS/Parc no 8.py (3.14.6)
File Edit Format Run Options Window Help
def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array)- i -1):
            if array[j] > array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp
data = [-2,45,0,11,-9]
bubbleSort(data)
print('Sort Array in Ascending Order:')
print(data)
print("Sapna_29")
```

Output:-

```
RESTART: C:/Users/info/OneDrive/Documents/Sapna DS/Parc no 8.py (3.14.6)
Sort Array in Ascending Order:
[-9, -2, 0, 11, 45]
Sapna_29
```

b. Insertion sort

Code:-

```
Parc no 8.py - C:/Users/info/OneDrive/Documents/Sapna DS/Parc no 8.py (3.14.b)
File Edit Format Run Options Window Help
def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key

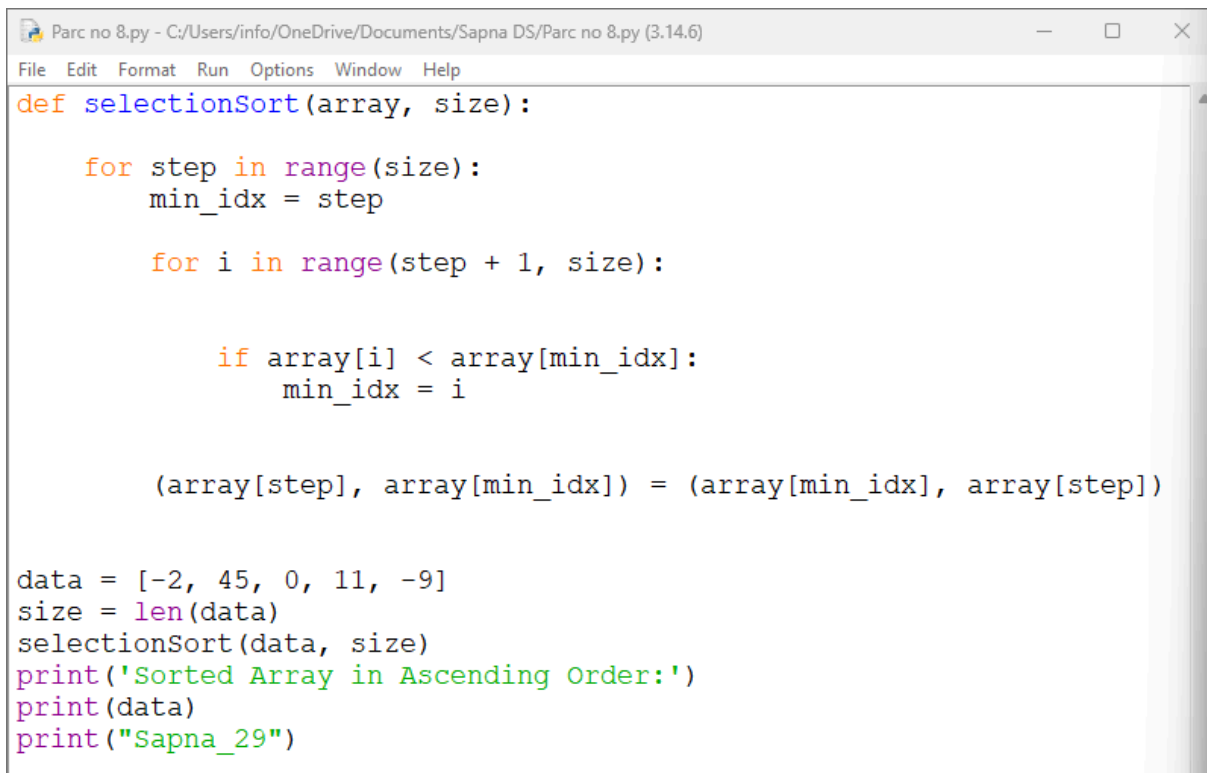
data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("Sapna_29")
```

Output:-

```
- RESTART: C:/Users/info/OneDrive/1
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
Sapna_29
```

c. Selection sort

Code:-

A screenshot of a Python IDE window titled "Parc no 8.py - C:/Users/info/OneDrive/Documents/Sapna DS/Parc no 8.py (3.14.6)". The window has a menu bar with "File", "Edit", "Format", "Run", "Options", "Window", and "Help". The code editor contains a Python script for selection sort. The script defines a function `selectionSort(array, size)` that sorts an array in ascending order. It uses nested loops: an outer loop for `step` from 0 to `size-1`, and an inner loop for `i` from `step+1` to `size-1`. The inner loop finds the minimum element and swaps it with the element at `step`. After the function definition, the script initializes `data = [-2, 45, 0, 11, -9]`, calculates `size = len(data)`, calls `selectionSort(data, size)`, and prints the sorted array, the original data, and a string "Sapna_29".

```
def selectionSort(array, size):  
  
    for step in range(size):  
        min_idx = step  
  
        for i in range(step + 1, size):  
  
            if array[i] < array[min_idx]:  
                min_idx = i  
  
        (array[step], array[min_idx]) = (array[min_idx], array[step])  
  
data = [-2, 45, 0, 11, -9]  
size = len(data)  
selectionSort(data, size)  
print('Sorted Array in Ascending Order:')  
print(data)  
print("Sapna_29")
```

OutPut:-

```
Sorted Array in Ascending Order:  
[-9, -2, 0, 11, 45]  
Sapna_29  
|
```